

Neural Networks Introduction

Neuron:

Neuron is (smallest unit or node) is the smallest block of neural networks. It takes inputs, x by weights, adds bias & applies activation function to decide output.

Perceptron:

Perceptron is basically one or more neurons combined to create a decision making machine.



Single Layered perceptron!

Limitations:

Works only for linearly separable data.

MultiLayer Perceptron finds the nonlinearity in data.

Input Layer Hidden Layers 1 Output Layer

Vanishing Gradient Problem:

This happens when the gradients becomes very small & There would be negligible updates in weights/bias so stuck in training (or) becomes stagnant.

Steps to reduce (or) avoid:

- (i) Batch Normalization
- (ii) Right Activation fn
- (iii) Gradient clipping
- (iv) use models (Gates) like LSTM or GRU

How to Identify:

- (i) model weights converges to 0
- (ii) Loss function fails to decrease significantly.
- (iii) Visualization Technique - (gradient histograms/norms)

Exploding Gradient Problem:

This is opposite of vanishing gradient problem.

Root cause lies in weights of the network, rather than choosing activation function. High weights leads to high derivatives causing significant changes in new weights and does not converges and can lead to oscillate around the local minima.

Solve · → Batch Normalization → Gradient Clipping
→ Weight Initialization → Kernel Constraint
(norm = 2.0)